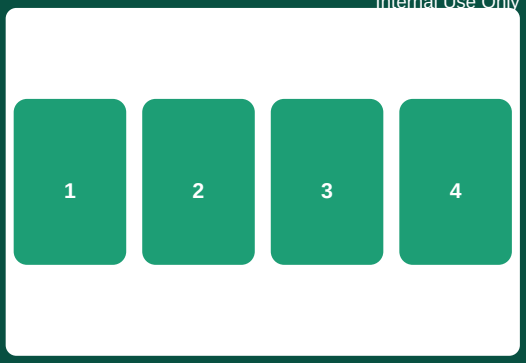
Internal Use Only

Design of Experiments

Systematically test combinations of input variables to find which factors most influence an output — and identify optimal



WHEN TO USE

When multiple factors may interact; when optimising a process or formulation; when one-factor-at-a-time testing is too slow or costly.

HOW TO APPLY — STEPS

1. Define the response variable (output) to be optimised
2. Identify input factors and levels to test
3. Choose experimental design (full factorial, fractional, Taguchi, RSM)
4. Run experiments in randomised order to avoid confounding
5. Analyse main effects and interactions
6. Confirm optimal settings with a separate validation run

PRACTITIONER TIP

! Randomise your run order without exception. Systematic patterns create confounding that invalidates all results.

KEY OUTPUTS

- Experimental design matrix
- Main effects and interaction plots
- Optimal factor settings
- Validation run confirmation

HOW TO MEASURE SUCCESS

% improvement in response variable; experimental efficiency vs one-factor-at-a-time

TOOL CATEGORY

Quality & Process